EduGenie

Requirements Analysis Phase

Software Project Requirements Document

Prepared for Academic Project / Innovation Activity

1. Introduction

The Requirements Analysis Phase identifies and documents the needs of students and other intended users of EduGenie. The purpose is to define what the system should provide, how users will interact with it, and the technical and quality requirements needed for successful development.

2. Objectives

The main objective of EduGenie is to provide an easy-to-use educational support platform that helps students understand subjects, clarify doubts, plan their studies, practise questions, and monitor their learning progress.

3. Stakeholders

The primary stakeholders are students, teachers or academic mentors, administrators, and the development team. Students are the main users, while teachers and administrators may support content management and academic monitoring.

4. Functional Requirements

- User registration and login.
- Student profile and learning preferences.
- Subject-wise educational support.
- Doubt clarification and concept explanations.
- Personalized study planning.
- Quiz and practice-question module.
- Progress tracking and performance summaries.
- Educational resource management.

5. Non-Functional Requirements

- Usability: The interface should be simple and student-friendly.
- Performance: Common user actions should respond quickly.
- Reliability: The system should operate consistently.
- Security: User information should be protected.
- Scalability: The system should support increasing users and content.
- Maintainability: The software should be easy to update and maintain.

6. User Requirements

User	Requirement
Student	Access learning support, ask doubts, practise questions, and track progress.
Teacher/Mentor	Support learning content and review student learning activities where applicable.
Administrator	Manage users, educational resources, and system settings.

7. Hardware and Software Requirements

- Hardware: Computer or mobile device with sufficient processing capability and internet connectivity.
- Operating system: Windows, Linux, Android, iOS, or another supported platform depending on implementation.
- Software: Suitable programming language, database system, development environment, and web/mobile framework.
- Optional AI/ML services may be integrated for intelligent educational assistance.

8. Data Requirements

EduGenie may store user profile information, learning preferences, study plans, quiz responses, progress information, and educational content. Appropriate access control and data protection should be considered during implementation.

9. Constraints

The system may depend on internet availability, device compatibility, data quality, available educational content, and the accuracy of automated responses. The project should also consider development time and available resources.

10. Expected Outcome

The requirements analysis should provide a clear foundation for designing and developing EduGenie. A well-defined set of requirements will help reduce ambiguity, guide implementation, and make it easier to evaluate whether the completed system meets student needs.

11. Conclusion

The Requirements Analysis Phase converts the initial EduGenie idea into clear and measurable system requirements. It establishes the expected functions, users, technical needs, quality attributes, and constraints that will guide the next stages of the project.